

Reference Schematics For RK3588S

RK3588S_Tablet_Demo_SCH

Main Functions Introduction

- 1) Charger: 1Cell Battery_QC
- 2) PMIC: 1 x RK806-1+DiscretePower
- 3) RAM: 2 x 32bits LPDDR4/4x
- 4) ROM: eMMC5.1(Default)
- 5) Support: 1 x Type-C 3.0(with DP function)
- 6) Support: 1 x 4Lanes MIPI D/CPHY RX Camera
- 7) Support: 1 x 2Lanes MIPI DPHY RX Camera
- 8) Support: 1 x 4Lanes MIPI D/CPHY TX
- 9) Support: a/b/g/n/ac/ax 2T2R WIFI(PCIE) + BT5.0
- 11) Support: 1 x Headphone + 2 x Speaker out + 1 x Analog MIC
- 12) Support: 2 x PDM MIC Array
- 13) Support: Gyroscope+G-sensor+Ambient Light+Proximity

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Note

The power suffix S0 or S3 means:

S3: Keep power On during sleeping

S0:Power off during sleeping

Generate Bill of Materials

Header:

Item\Part\Description\PCB Footprint\Reference\Quantity\Option

Combined property string:

{Item}\t{Value}\t{Description}\t{PCB Footprint}\t{Reference}\t{Quantity}\t{Option}

Description

Note

Option

Notes

NOTE 1:

Component parameter description

1. DNP stands for component not mounted temporarily

2. If Value or option is DNP, which means the area is reserved without being mounted

NOTE 2:

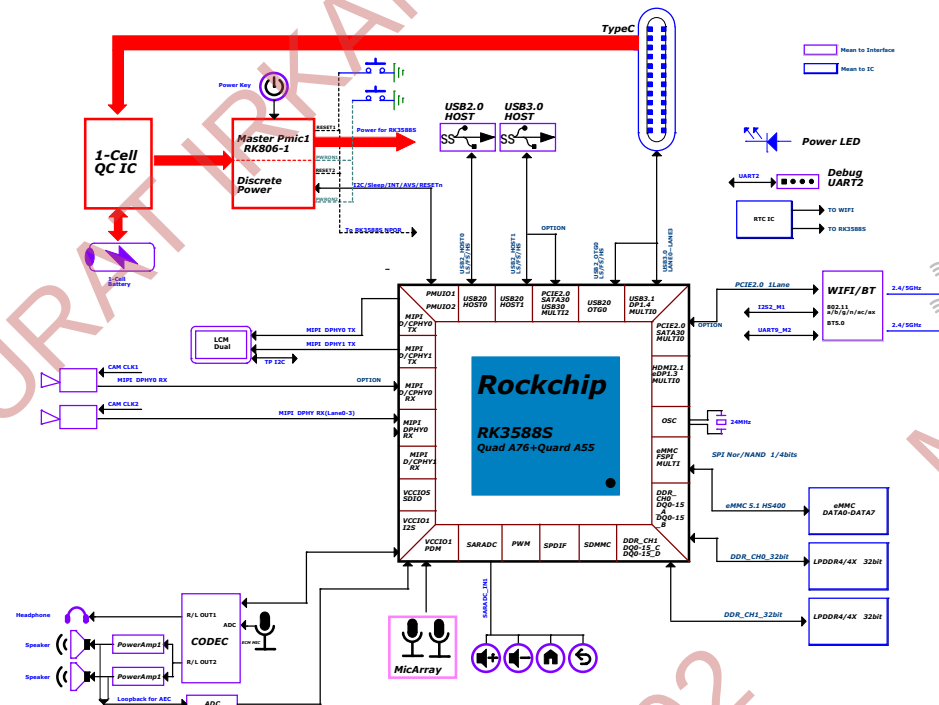
Please use our recommended components to avoid too many changes.
For more informations about the second source,please refer to our AVL.

Revision History

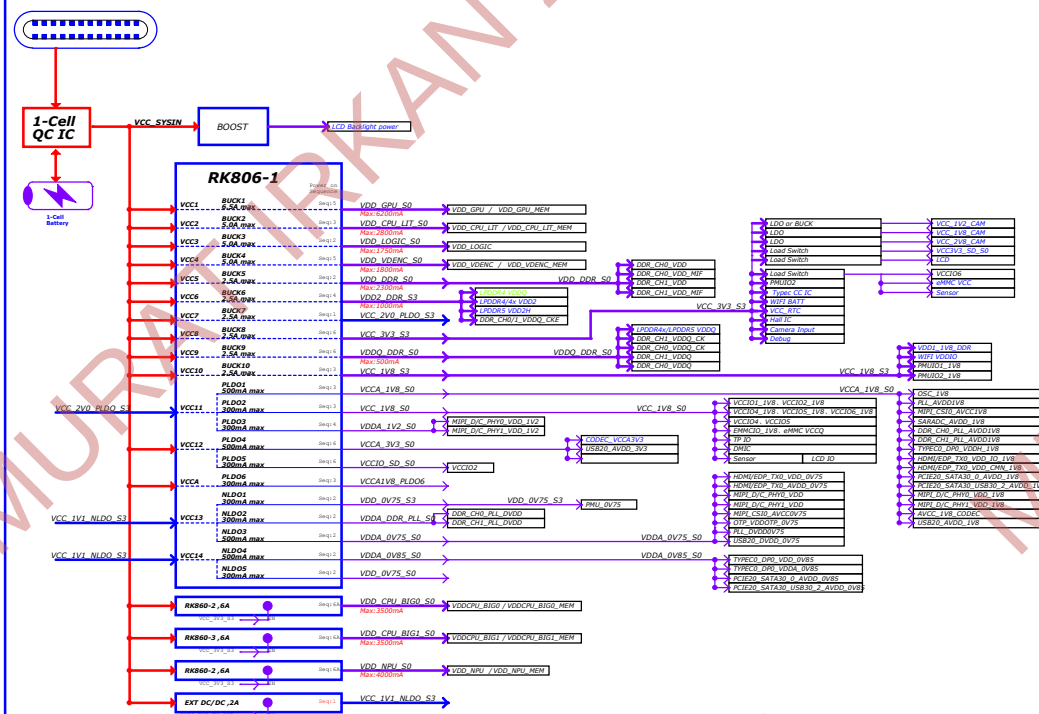
Version	Date	By	Change Dscription	Approved
V1.0	2022-01-05	Joseph.Wei	1:Revision preliminary version	
V1.1	2022-02-18	Joseph.Wei	1.C1604,C1612的电容改成1uF/4V。 2.为了减少待机功耗，将PMUIO2电源域改成1.8V，此IO域对应外设IO电压相应修改 3.把L2203，L2205，L2207，L2300，L2301，L2302电感由0.22uH(TDK)改为0.24uH(Sunlord); L2201的电感由0.22uH(TDK)改为0.22uH (Sunlord)，封装IND_404020。	



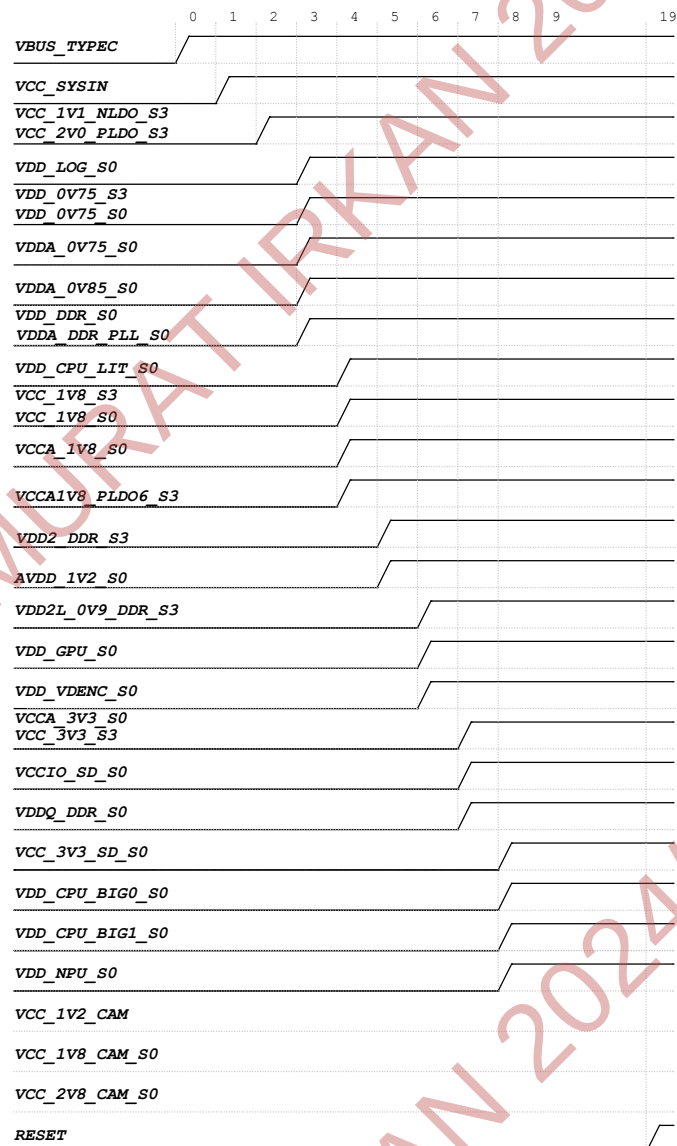
RK3588S Tablet Demo Block Diagram for 1-Cell Charger



Power tree for 1-Cell Charger



Power Sequence

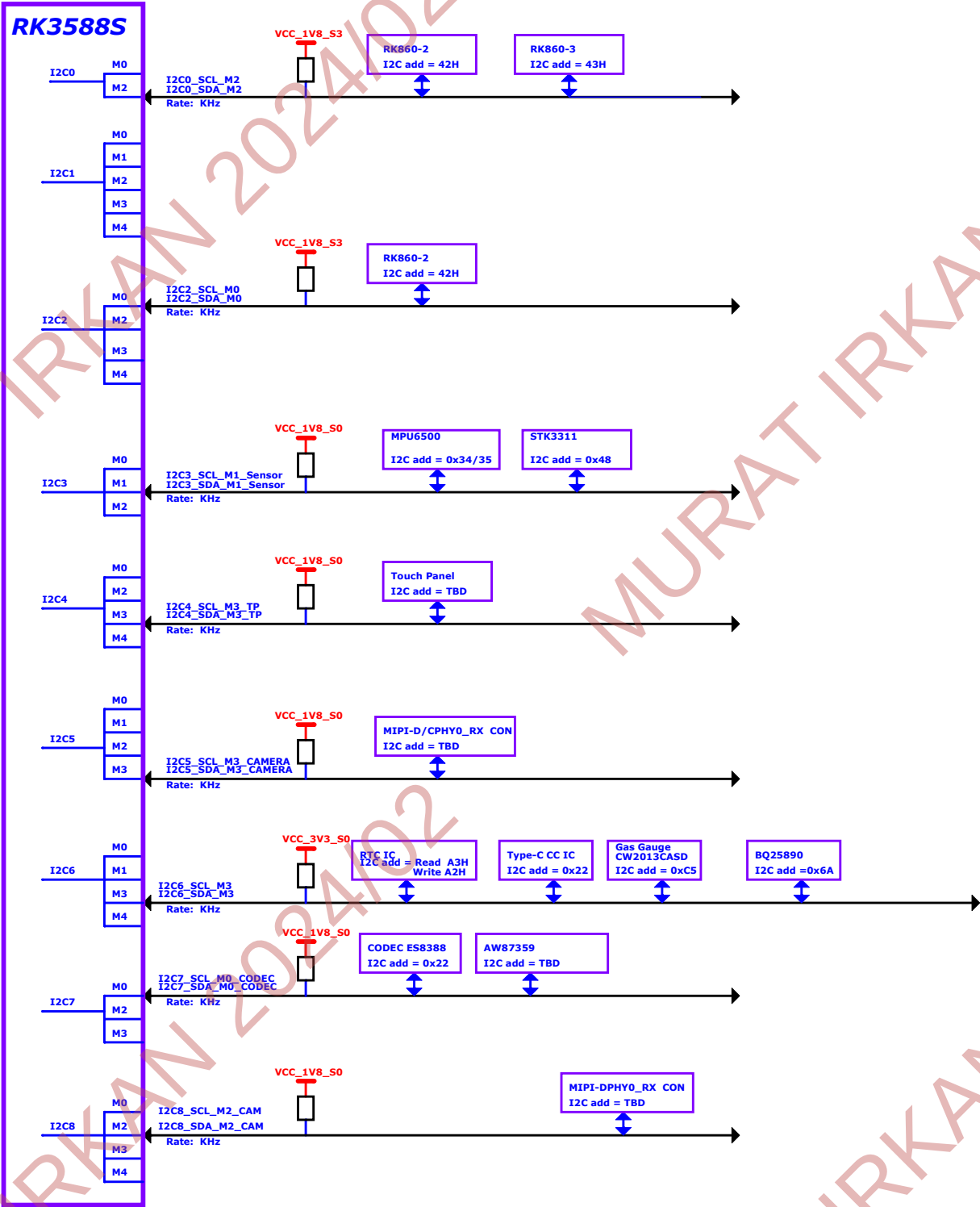


Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Sleep ON/OFF	Peak Current	Sleep Current
VCC_SYSIN	RK806-1_BUCK1	6.5A	VDD_GPU_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK2	5A	VDD_CPU_LIT_S0	Slot:3	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK3	5A	VDD_LOG_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK4	3A	VDD_VDENC_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK5	2.5A	VDD_DDR_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK6	2.5A	VDD2_DDR_S3	Slot:4	ADJ FB=0.5V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK7	2.5A	VCC_2V0_PLDO_S3	Slot:1	2.0V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK8	2.5A	VCC_3V3_S3	Slot:6	3.3V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK9	2.5A	VDDQ_DDR_S0	Slot:6	ADJ FB=0.5V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK10	2.5A	VCC_1V8_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_2V0_PLDO_S3	RK806-1_PLDO1	0.5A	VCC_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO2	0.3A	VCCA_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO3	0.3A	VDDA_1V2_S0	Slot:4	1.2V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_PLDO4	0.5A	VCCA_3V3_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO5	0.3A	VCCIO_SD_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO6	0.3A	VCCA1V8_PLDO6_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_1V1_NLDO_S3	RK806-1_NLDO1	0.3A	VDD_0V75_S3	Slot:2	0.75V	ON	ON	TBD	TBD
	RK806-1_NLDO2	0.3A	VDDA_DDR_PLL_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO3	0.5A	VDDA_0V75_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_1V1_NLDO_S3	RK806-1_NLDO4	0.5A	VDDA_0V85_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO5	0.3A	VDD_0V75_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-2	6A	VDD_CPU_BIG0_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-3	6A	VDD_CPU_BIG1_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-2	6A	VDD_NPU_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	EXT BUCK	2A	VCC_1V1_NLDO_S3	Slot:1	1.1V	ON	ON	TBD	TBD
VCC_SYSIN	EXT BUCK	2A	VDD2L_0V9_DDR_S3	Slot:5	0.9V	ON	ON	TBD	TBD
VCC_SYSIN	EXT BUCK	2.5A	VCC_3V3_SD_S0	Slot:6A	3.3V	ON	OFF	TBD	TBD
VCC_SYSIN	EXT_BUCK or LDO	2A	VCC_1V2_CAM_S0	OFF	1.2V	OFF	OFF	TBD	TBD
VCC_SYSIN	LDO	0.5A	VCC_1V8_CAM_S0	OFF	1.8V	OFF	OFF	TBD	TBD
VCC_SYSIN	LDO	0.5A	VCC_2V8_CAM_S0	OFF	2.8V	OFF	OFF	TBD	TBD

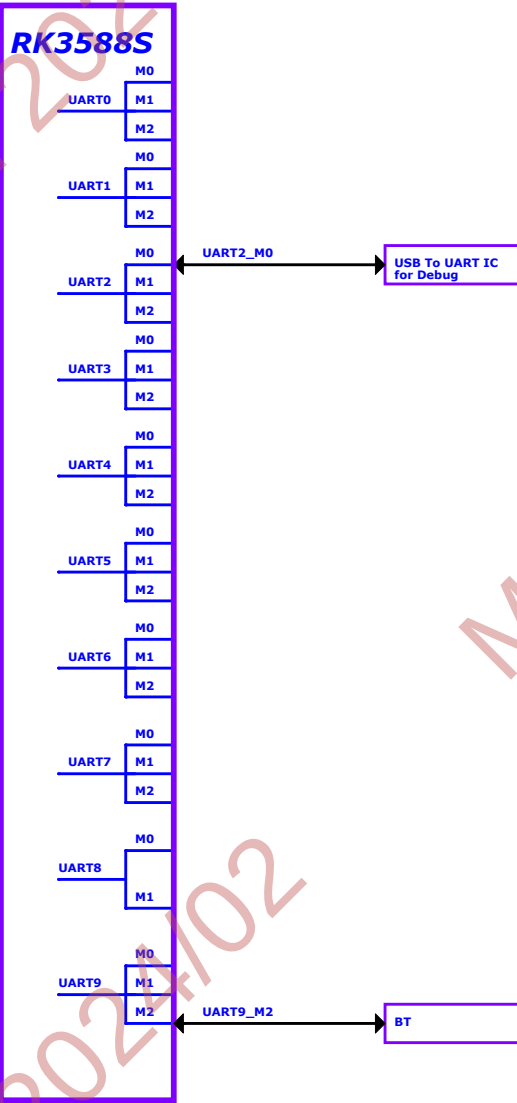
IO Power Domain Map

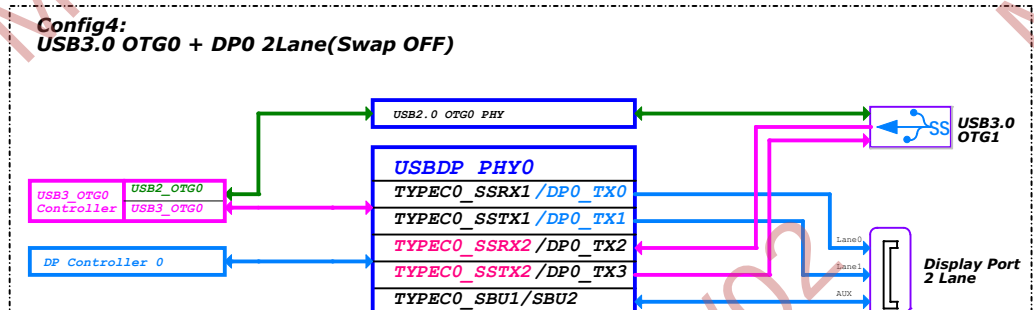
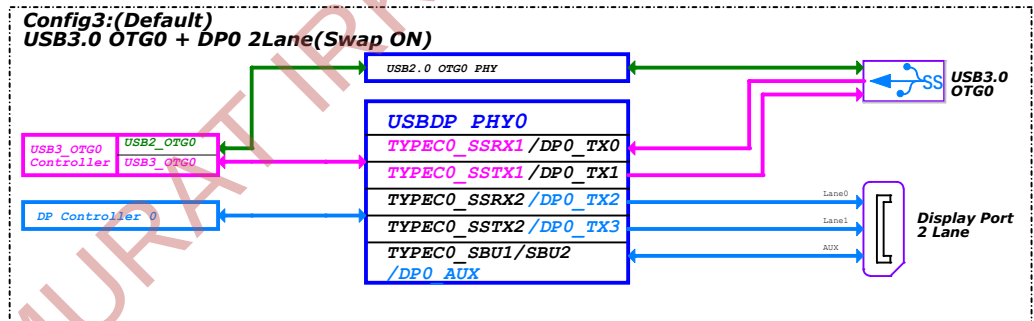
IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	Operating Voltage
PMUIO1	Pin N36 N37	1.8V Only	PMUIO1_1V8	VCC_1V8_S3	1.8V
PMUIO2	Pin V37 Y37	1.8V or 3.3V	PMUIO2_1V8	VCC_1V8_S3	1.8V
EMMCIO	Pin V35 V36	1.8V Only	PMUIO2	VCC_1V8_S3	1.8V
	Pin AC35	1.8V Only	EMMCIO_1V8	VCC_1V8_S0	1.8V
	Pin AC36	1.8V Only			
VCCIO1	Pin H31	1.8V Only	VCCIO1_1V8	VCC_1V8_S0	1.8V
VCCIO2	Pin AK11	1.8V or 3.3V	VCCIO2_1V8	VCC_1V8_S0	1.8V
	Pin AK10	1.8V or 3.3V	VCCIO2	VCC_1V8_S0	1.8V
VCCIO4	Pin G27 G28	1.8V or 3.3V	VCCIO4_1V8	VCC_1V8_S0	1.8V
	Pin G31	1.8V or 3.3V	VCCIO4	VCC_1V8_S0	1.8V
VCCIO5	Pin AF35 AF36	1.8V or 3.3V	VCCIO5_1V8	VCC_1V8_S0	1.8V
	Pin AC33 AC34	1.8V or 3.3V	VCCIO5	VCC_1V8_S0	1.8V
VCCIO6	Pin A134	1.8V or 3.3V	VCCIO6_1V8	VCC_1V8_S0	1.8V
	Pin A133 AM33	1.8V or 3.3V	VCCIO6	VCC_1V8_S0	1.8V

I2C MAP

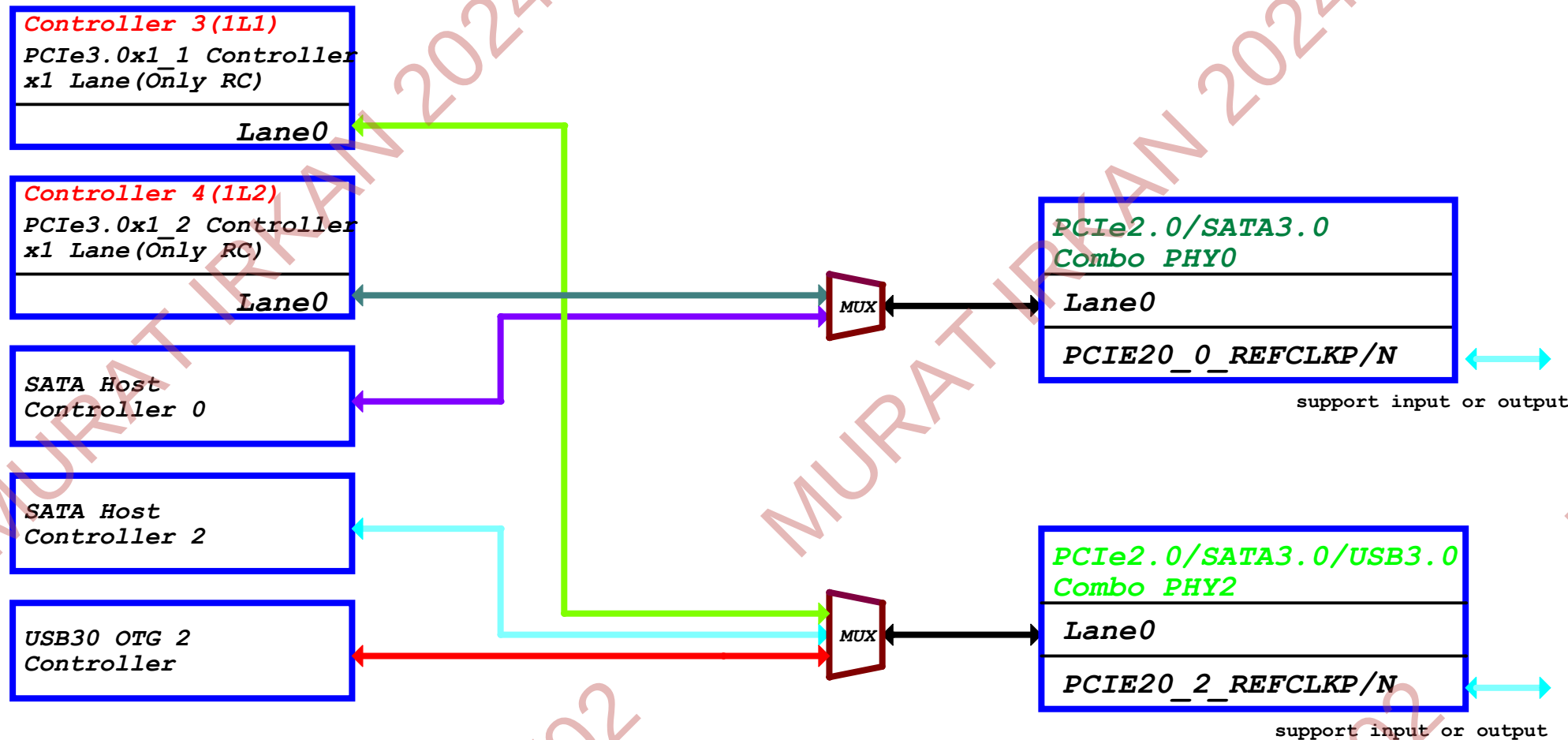


UART MAP



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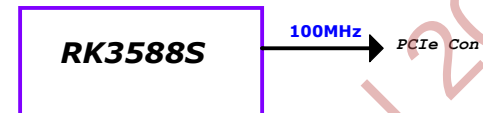
PCIe/SATA Connector Diagram



PCIe Controller Configure Table

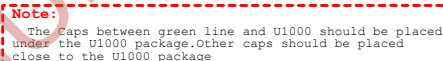
Controller Name	Data & Clk Lane Configure		Control GPIO
	CLK LANE	DATA LANE	
PCIE20X1_1 RC	PCIE20_2_REFCLKP PCIE20_2_REFCLKN	PCIE20_2_TX PCIE20_2_RX	PCIE20X1_1_CLKREQ M* PCIE20X1_1_WAKEN M* PCIE20X1_1_PERSTN M* PCIE20X1_1_BUTTON_RSTN
PCIE20X1_2 RC	PCIE20_0_REFCLKP PCIE20_0_REFCLKN	PCIE20_0_TX PCIE20_0_RX	PCIE20X1_2_CLKREQ M* PCIE20X1_2_WAKEN M* PCIE20X1_2_PERSTN M* PCIE20X1_2_BUTTON_RSTN

PCIe2.0 REFCLK



Note:
PCIE20_*_REFCLKP/N is output or input gpio
M*=Mean to M0 or M1 or M2,it's the same source,Just multiplex to M0 or M1 or M2,Only use one at the same time.

RK3588S (Power&Gnd)



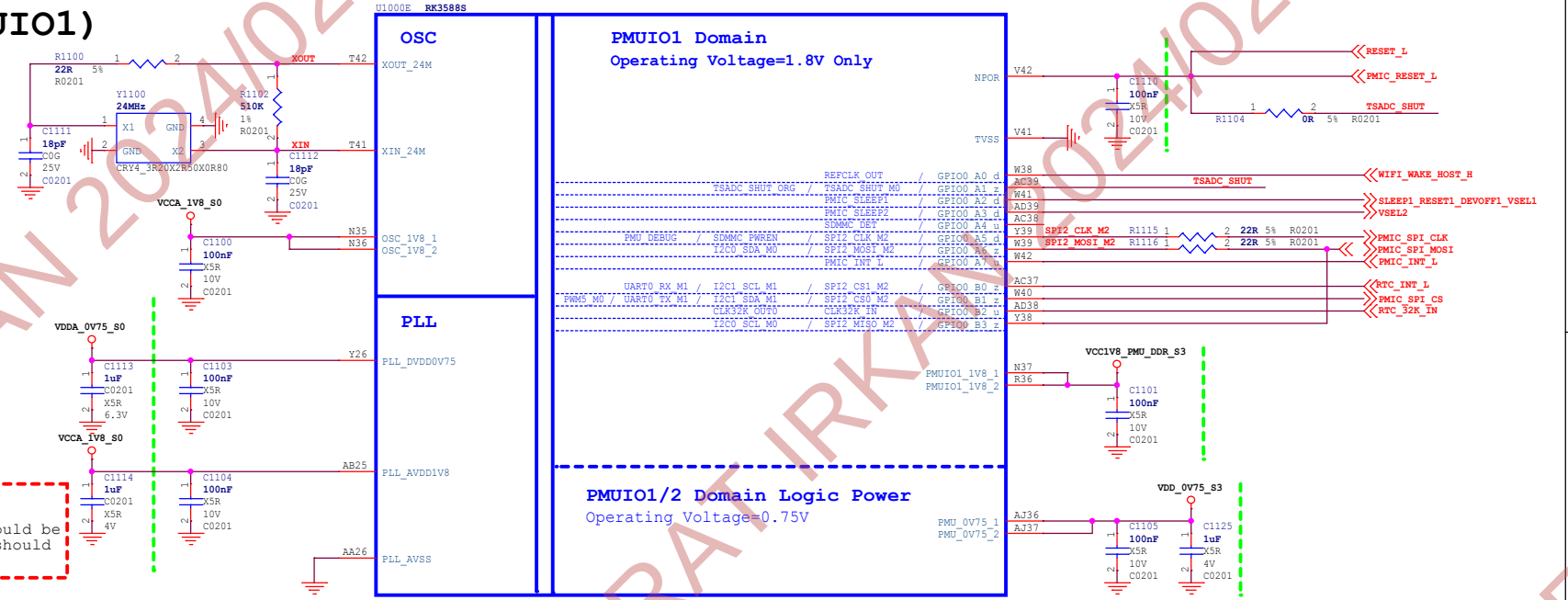
RK3588S (OSC/PLL/PMUIO1)

Note:
Adjusted the load capacitance according to the crystal specification

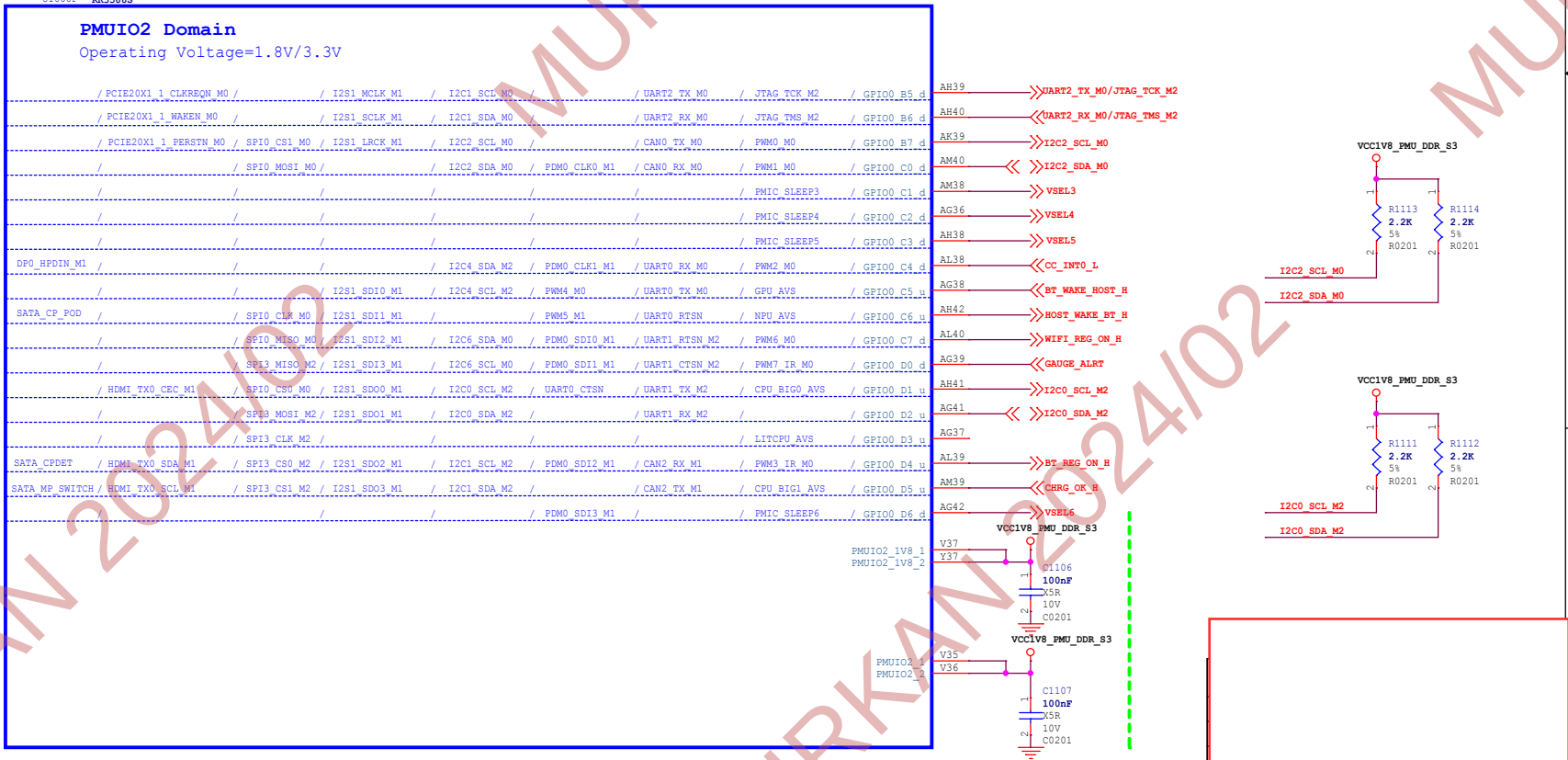
The CL is the load capacitance of the crystal that is recommended by the crystal vendors to obtain target clock frequency.

$CL = \{CL1 * CL2 / (CL1 + CL2)\} + PCB \text{ strays}$
Total CL<=12pF

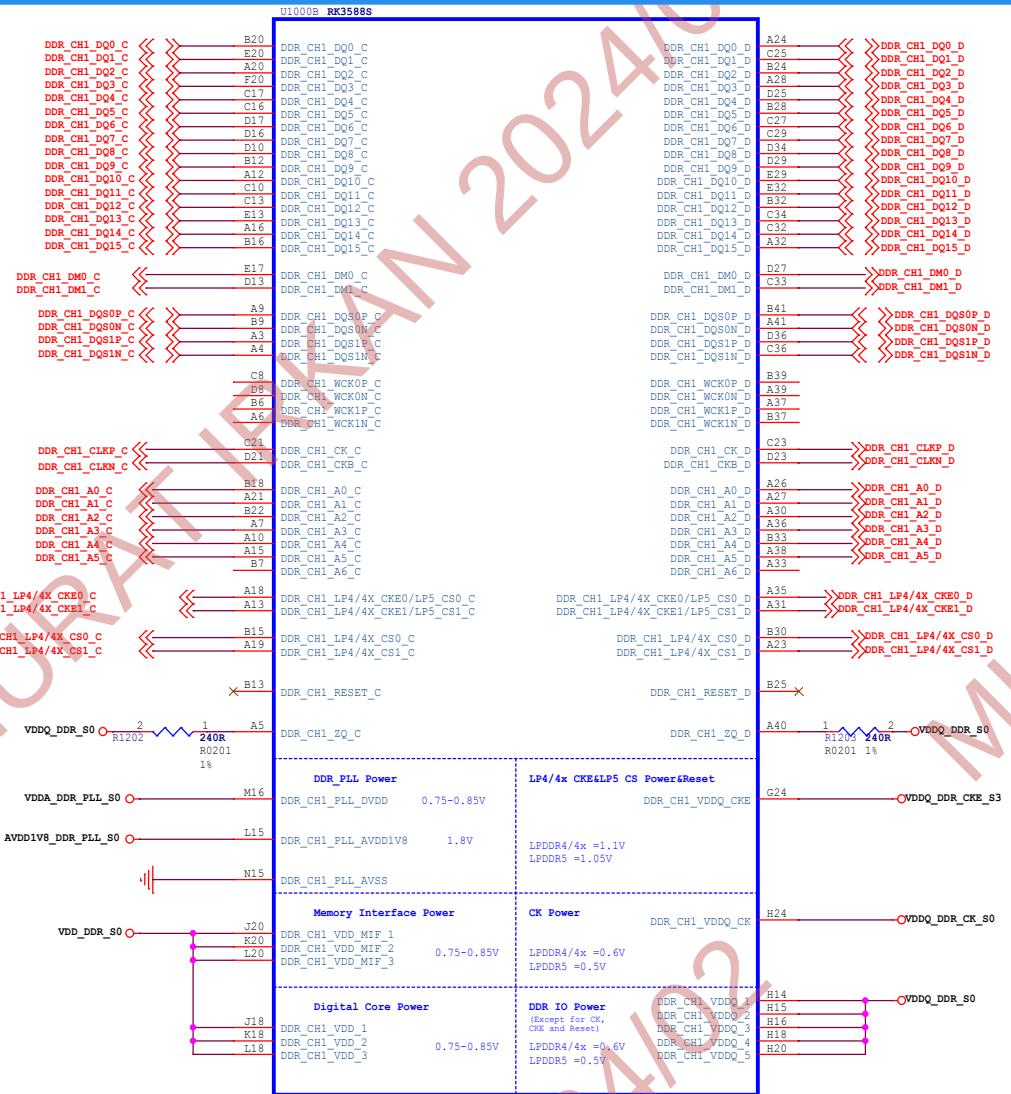
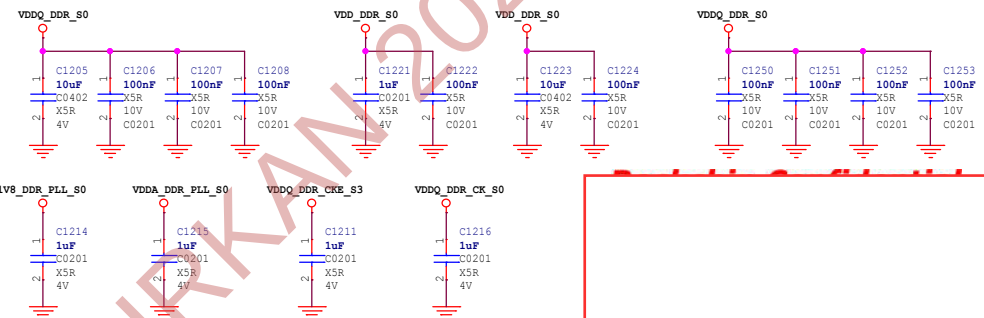
Note:
The Caps between green line and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package



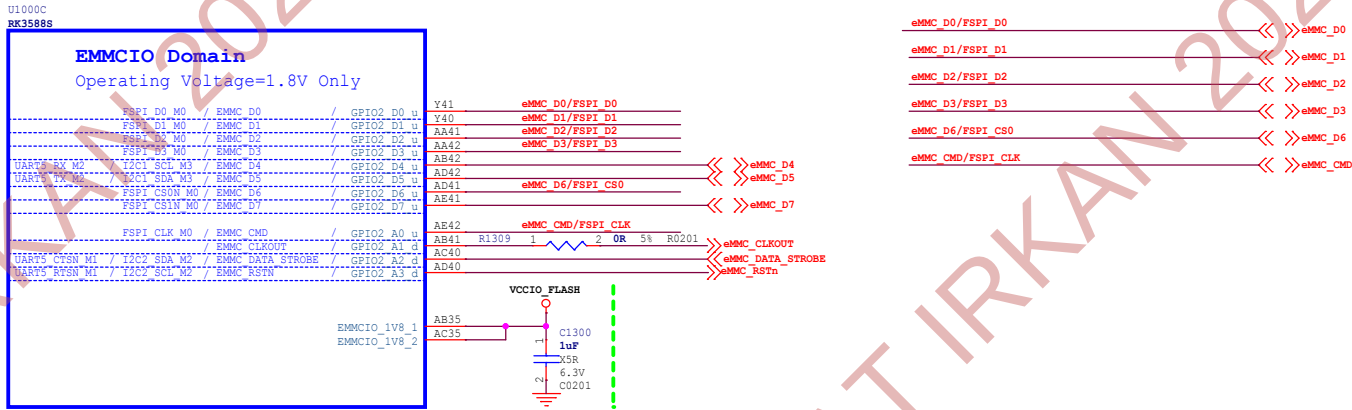
RK3588S (PMUIO2)



LPDDR4-MICRON 32GB

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RK3588S (EMMCIO Domain)



RK3588S (VCCIO2 Domain)



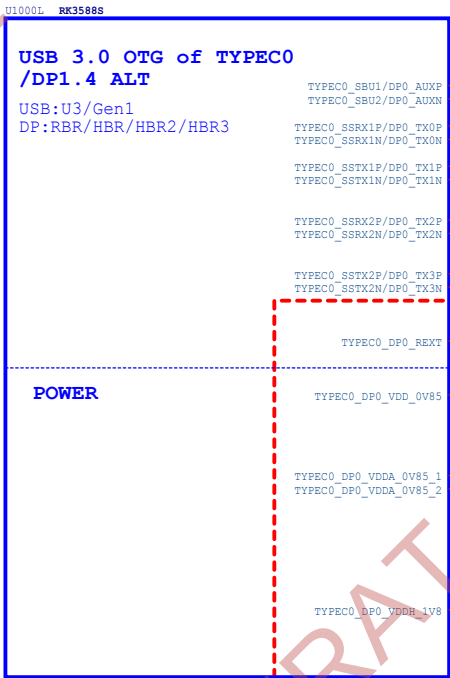
Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

RK3588S (USB3.0/DP1.4)

USB30/DP1.4 Alt Mode Configuration

Option1	DP x4Lane	DP_TX_Lane0-3
Option2	TYPEC x4Lane	SSTX 1P/1N SSTX 2P/2N SSRX 1P/1N SSRX 2P/2N
Option3	USB30X2Lane+DPX2Lane	USB30:SSTX 1P/1N SSRX 1P/1N DP:Lane2 Lane3
Option4	USB30X2Lane+DPX2Lane	USB30:SSTX 2P/2N SSRX 2P/2N DP:Lane0 Lane1

DP Lane
Swap Off:
Lane0/1/2/3_TXdata mapping to Lane0/1/2/3_TXDP/N
Swap On:
Lane0/1/2/3_TXdata mapping to Lane2/3/0/1_TXDP/N

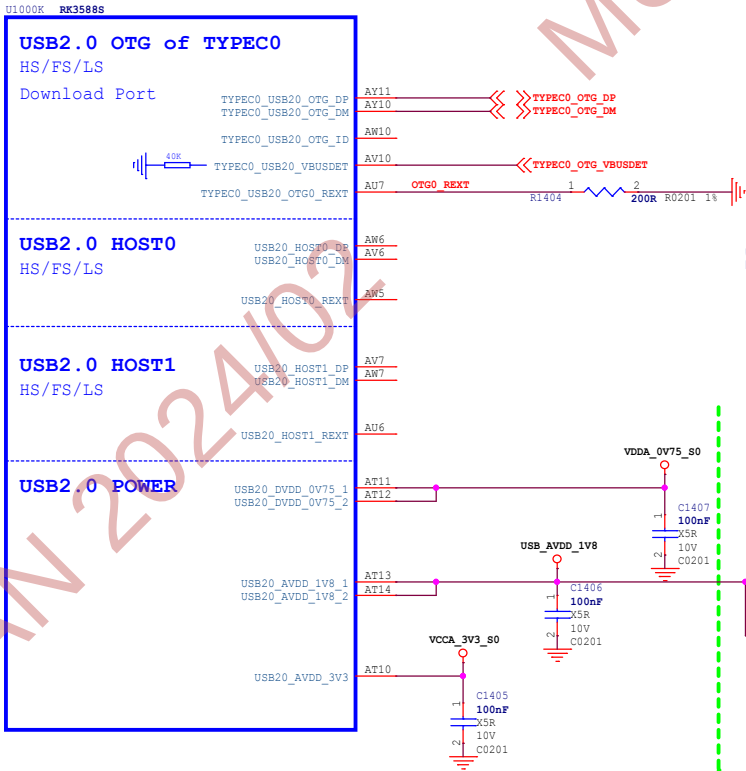


TYPEC&DP MUX Differential Pair:
DATE:95 Ohm +10%
For Typec

USB30 Differential Pair: DATE:90 Ohm +10%
DP Differential Pair: DATE:100 Ohm +10%
For USB30 For DP

Do not delete!!!
If TYPECO is not used:
Signal:leave floating
REXT:8.2K ohm 1% resistor must
be connected externally
Power: Must supply power

RK3588S (USB2.0)



Note:
The USB20_VBUSDET pin internal has a pull-down resistance(40K ohm) to ground,The resistance creates a voltage with the external series 24K ohm resistor.The VBUSDETpin voltage range <=3.3V.

Note:
TYPECO_USB20_OTG:
DP/DM:Must used for download
ID:According to demand,if not used,Leave floating
VBUSDET:Must provide
REXT:200ohm 1% resistor must be connected externally
Power: Must supply power

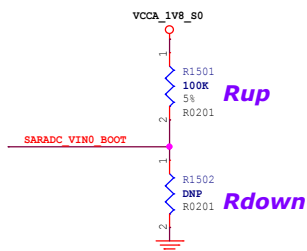
USB20_HOST0/USB20_HOST1:
If not used:
DP/DM:Leave floating
REXT:Leave floating

Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

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———>> SARADC_VIN1_KEY/RECOVERY
———>> SARADC_VIN2_LCD_ID
———>> SARADC_VIN4_BATT_TC_L

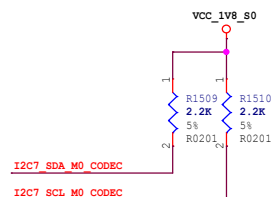
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Item	Rup	Rdown	ADC	BOOT MODE(saradc_in5)
LEVEL1	DNP	100K	0	USB (Maskrom mode)
LEVEL2	100K	20K	682	SD Card-USB
LEVEL3	100K	51K	1365	EMMC-USB
LEVEL4	100K	100K	2047	FSPI M0-USB
LEVEL5	100K	200K	2730	FSPI M1-USB
LEVEL6	100K	499K	3412	FSPI M2-USB
LEVEL7	100K	DNP	4095	FSPI_M2-FSPI_M0-EMMC -SD Card-USB

U1000G RK3588S

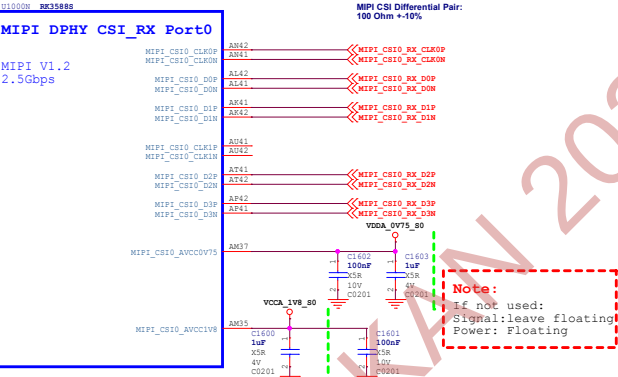
VCCIO1 Domain
Operating Voltage=1.8V Only



Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

RK3588S(MIPI_DPHY CSI0 RX)

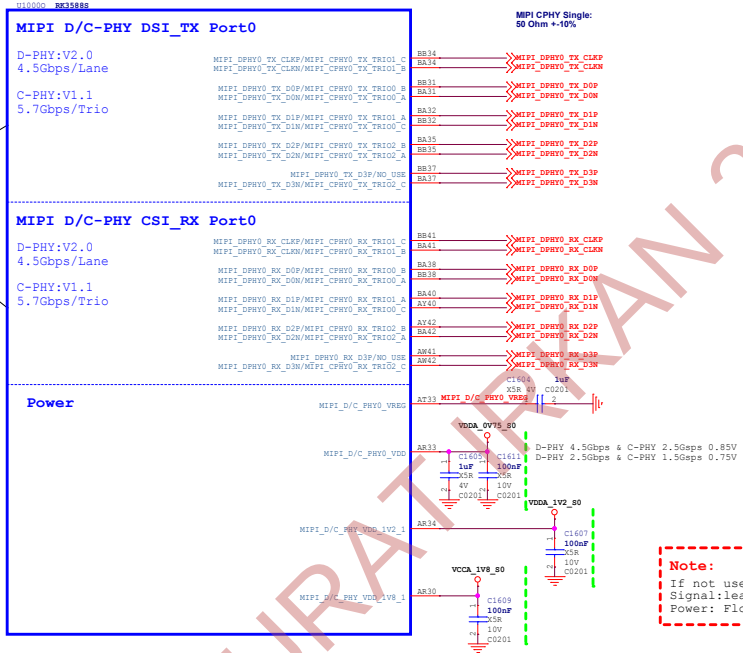


Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0 MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

Note:
When in single clock lane mode, CLK0P/0N is the clock lane from Data lane0 to Data lane3, but clock lane1 is invalid; In dual clock lanes mode, CLK0P/0N is the clock lane of Data lane0 and Data lane1, while CLK1P/1N is the clock lane of Data lane2 and Data lane3.

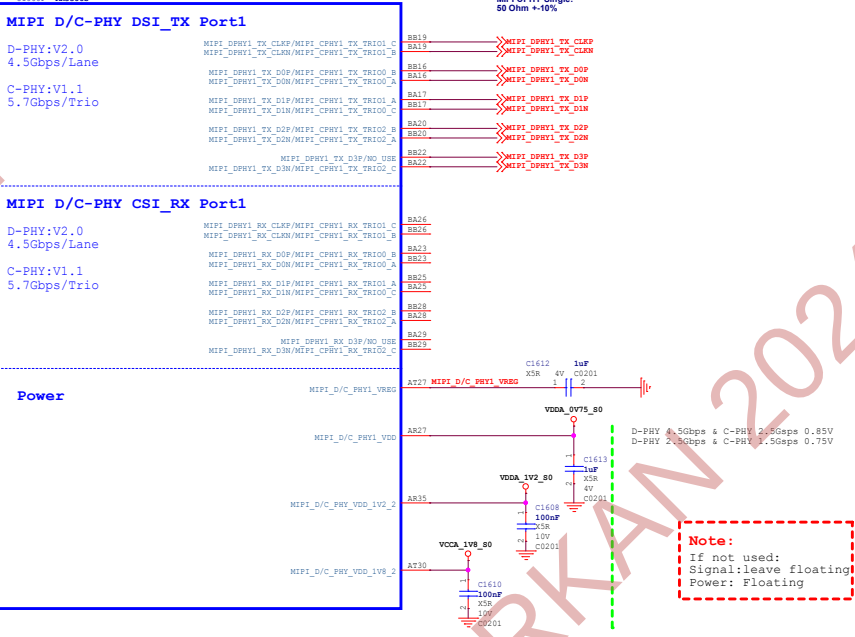
Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

RK3588S(MIPI_D/C PHY0)



TX and RX port must work in the same mode, DPHY or CPHY

RK3588S(MIPI_D/C PHY1)



TX and RX port must work in the same mode DPHY or CPHY

Note:
If not used:
Signal:leave floating
Power: Floating

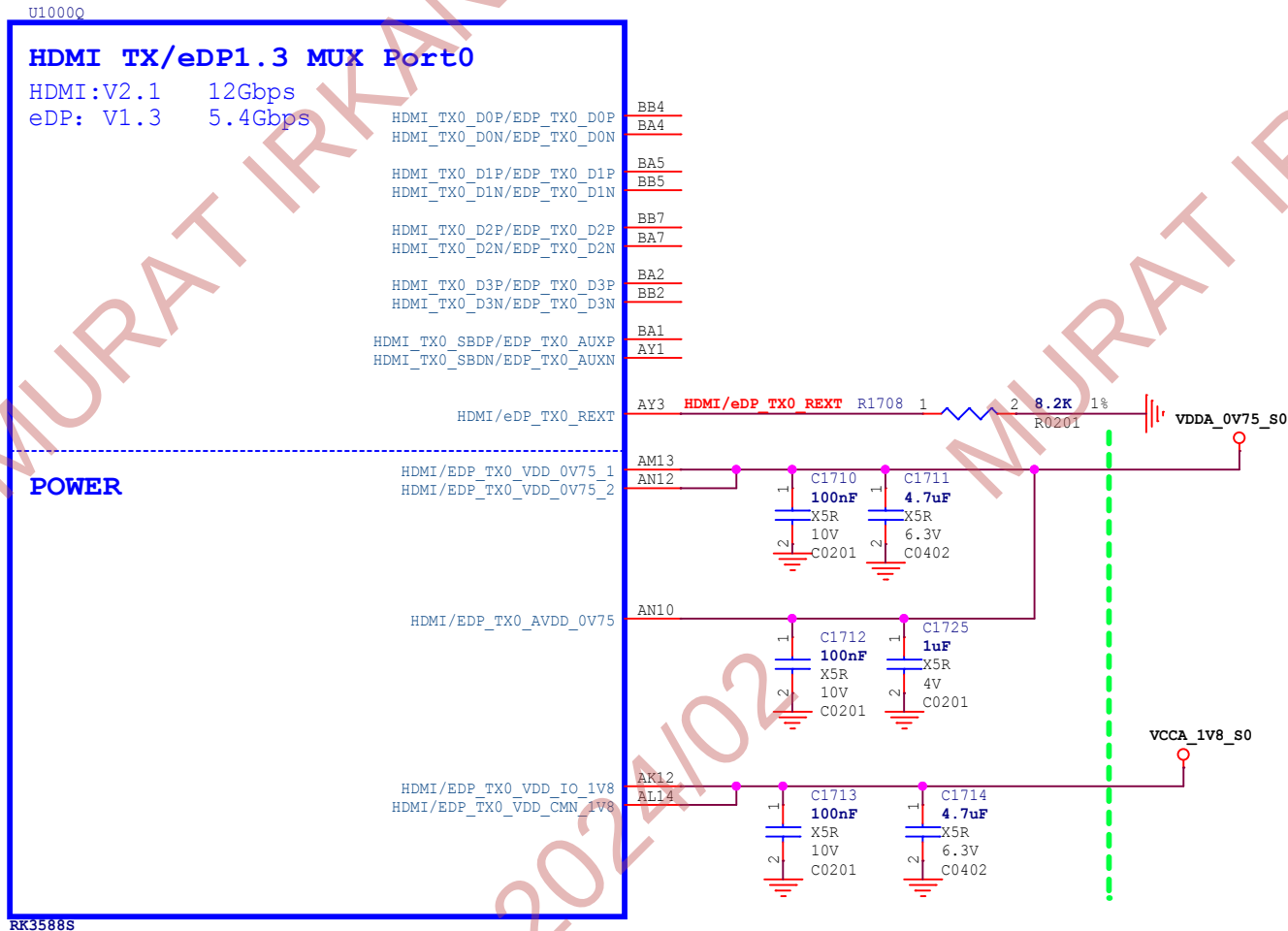
RK3588S (HDMI2.1 TX/eDP1.3 TX)

Note:

The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.

eDP TX
100 Ohm $\pm 10\%$

HDMI TX
100 Ohm $\pm 10\%$



Note:

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If not used:
Signal:leave floating
Power: Floating or tie to VSS
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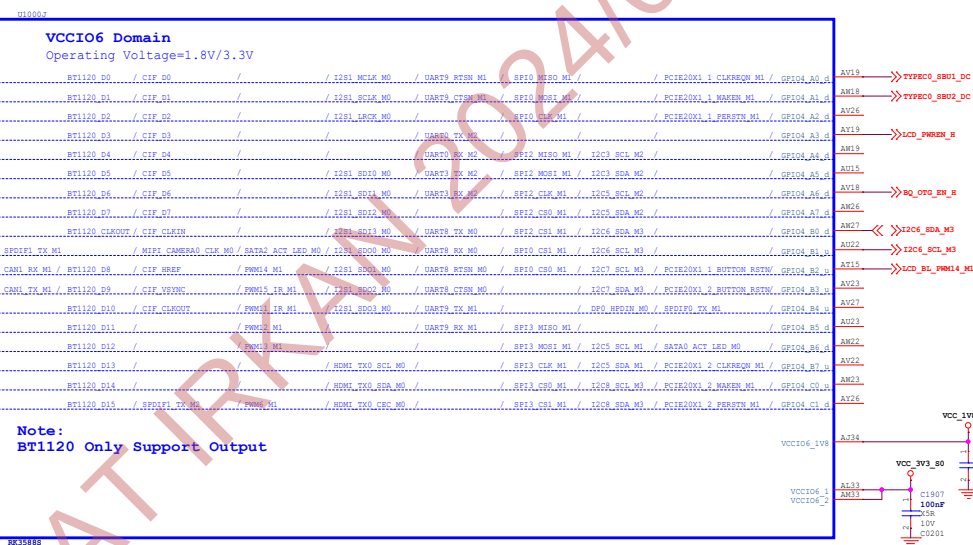
! Note:

Caps of between dashed green lines and U1000
should be placed under the U1000 package

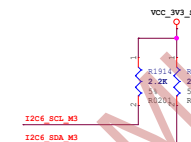
Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package

Controller Name	Data & Clk Lane Configure		Control GPIO
	CLK LANE	DATA LANE	
PCIE20X1_1 RC	PCIE20_2_REFCLKP PCIE20_2_REFCLKN	PCIE20_2_TX PCIE20_2_RX	PCIE20X1_1_CLKREQ_M* PCIE20X1_1_WAKEN_M* PCIE20X1_1_PERSTN_M* PCIE20X1_1_BUTTON_RSTN
PCIE20X1_2 RC	PCIE20_0_REFCLKP PCIE20_0_REFCLKN	PCIE20_0_TX PCIE20_0_RX	PCIE20X1_2_CLKREQ_M* PCIE20X1_2_WAKEN_M* PCIE20X1_2_PERSTN_M* PCIE20X1_2_BUTTON_RSTN

RK3588S (VCCIO6 Domain)

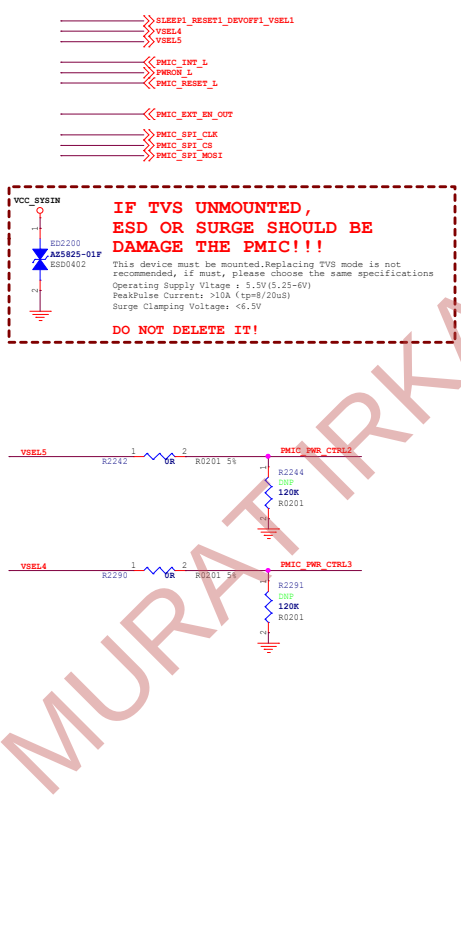


Note:
BT1120 Only Support Output

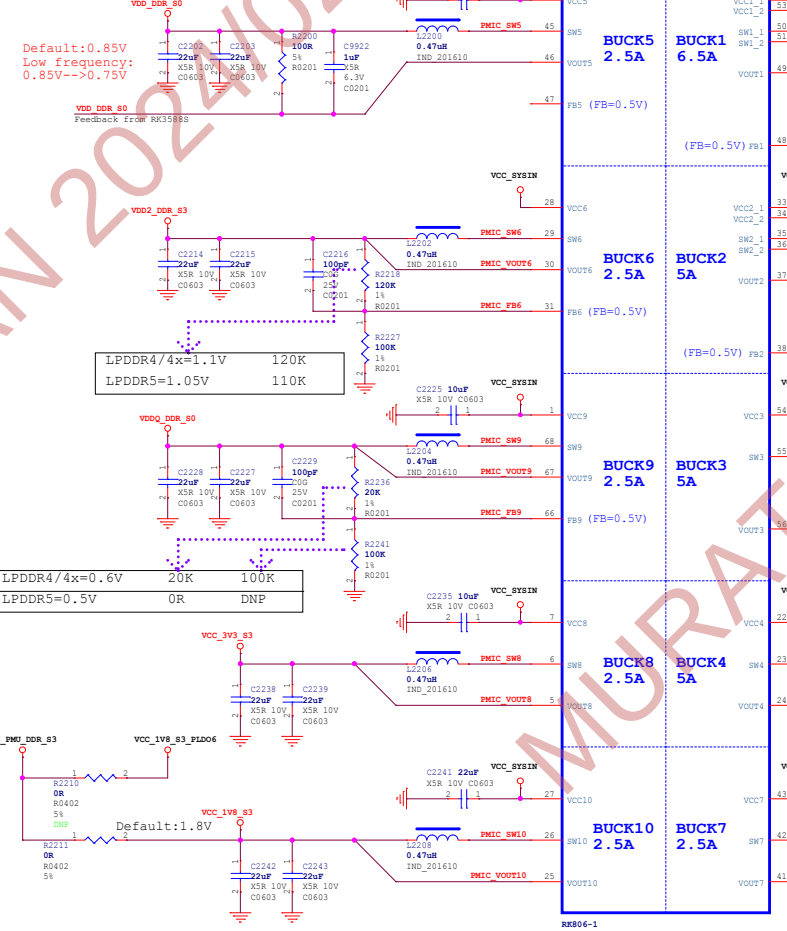


Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package

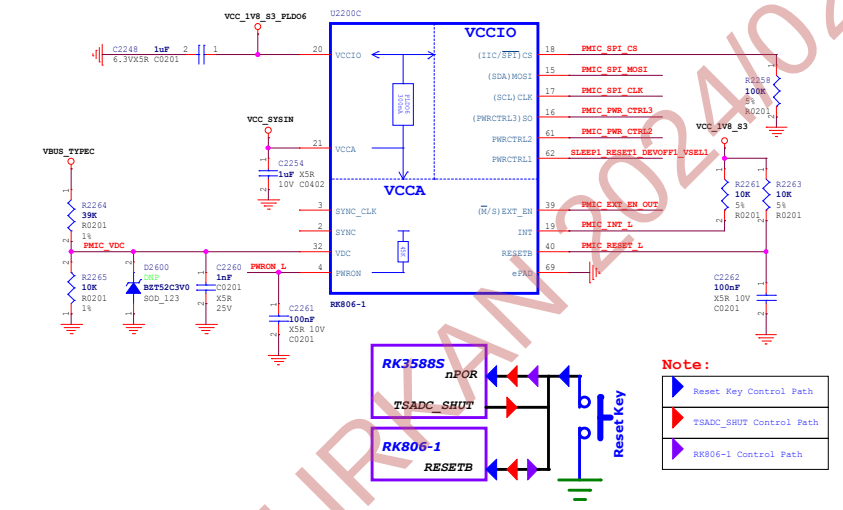
PMIC1 RK806-1



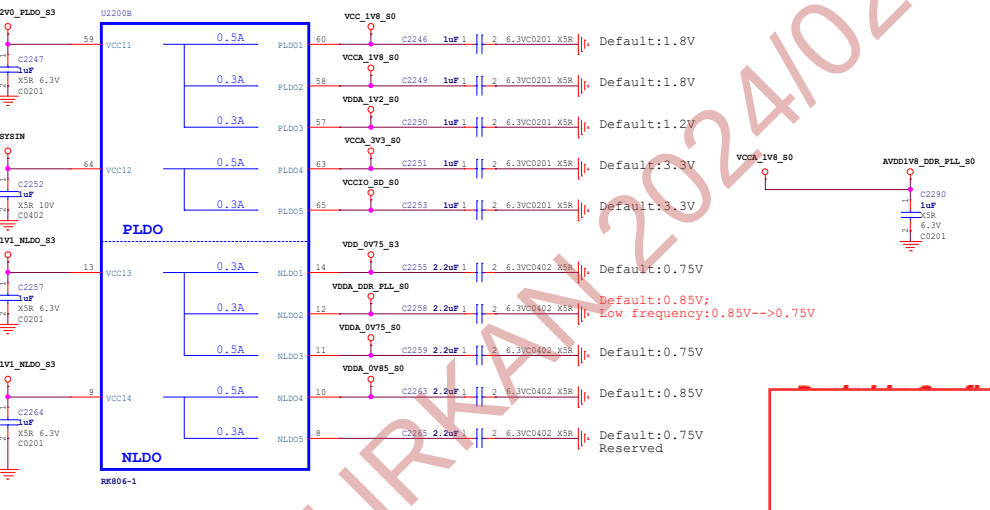
PMIC RK806-1 BUCK



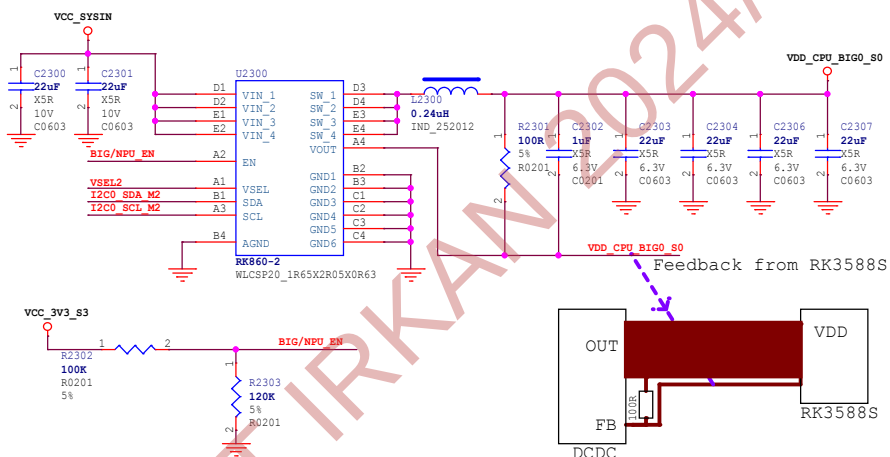
PMIC RK806-1 Management



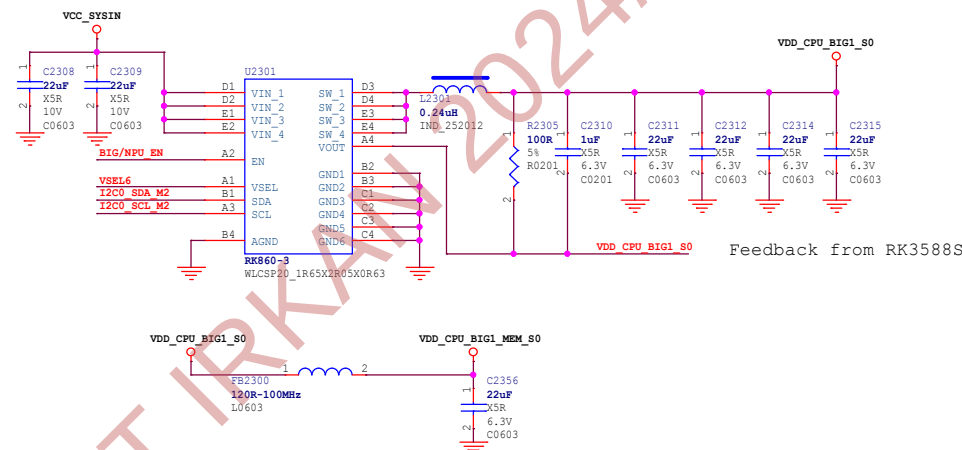
PMIC RK806-1 LDO



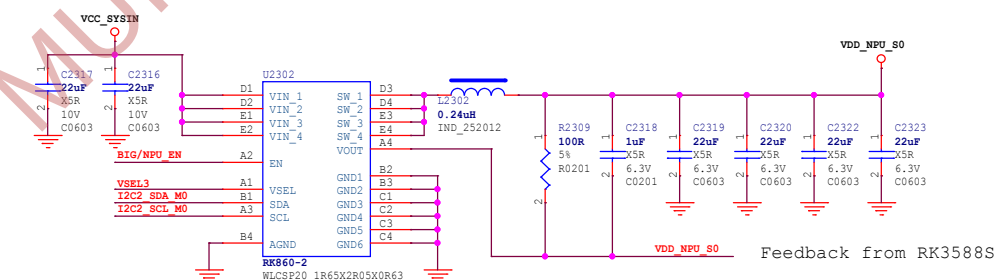
VDD_CPU_BIG0



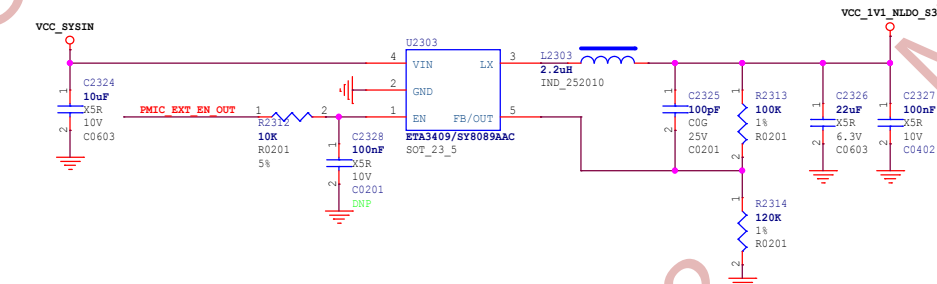
VDD_CPU_BIG1



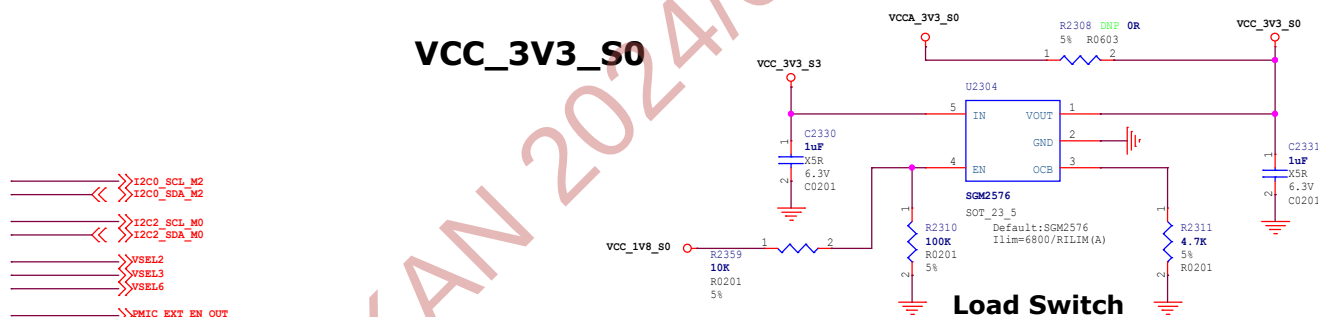
VDD_NPU



VCC_1V1_NLDO



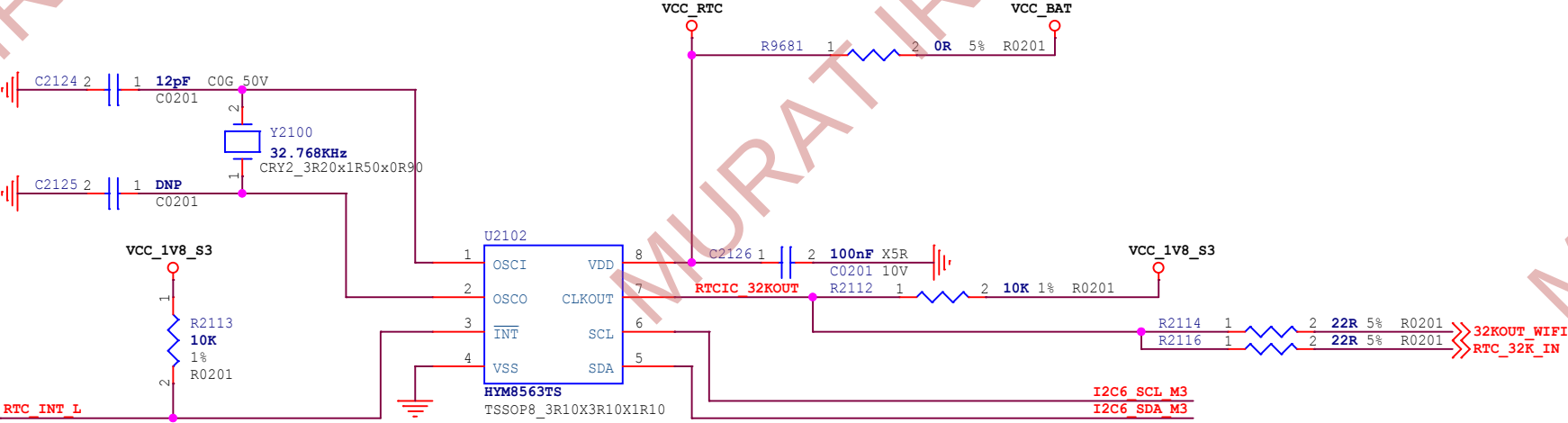
VCC_3V3_S0



Load Switch

RTC IC

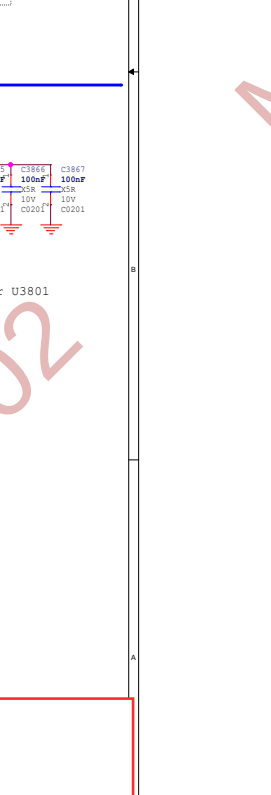
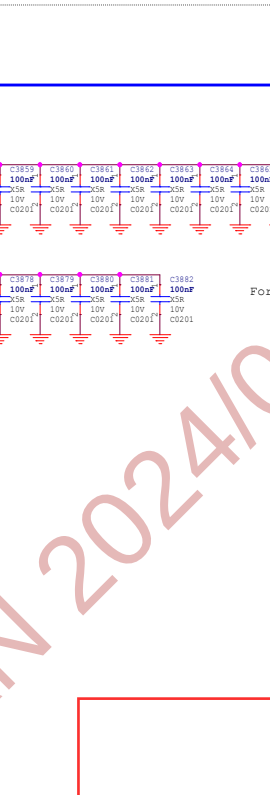
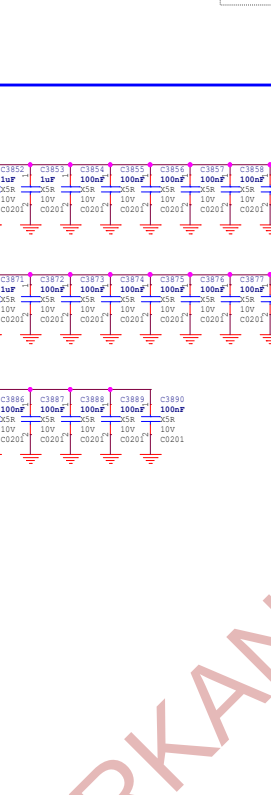
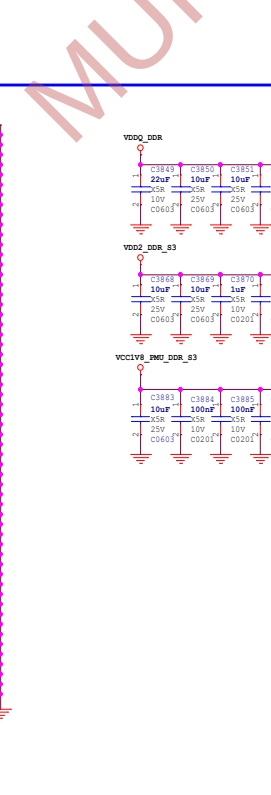
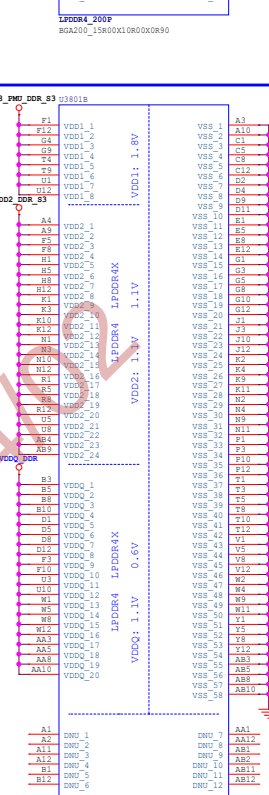
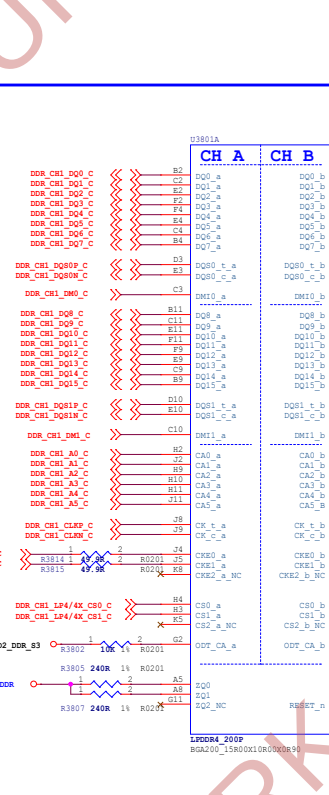
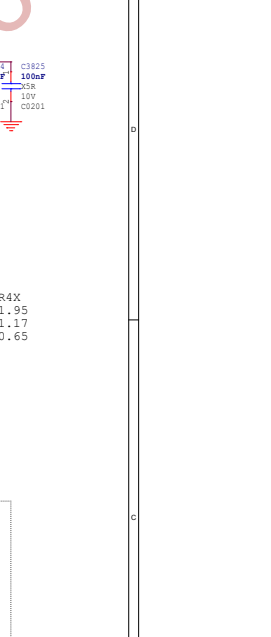
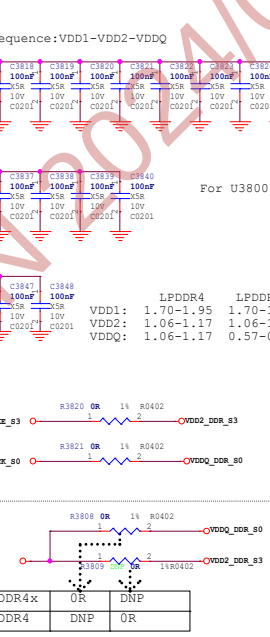
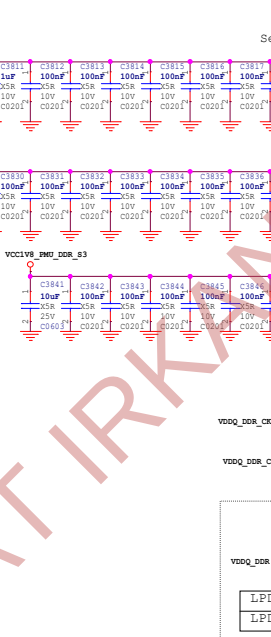
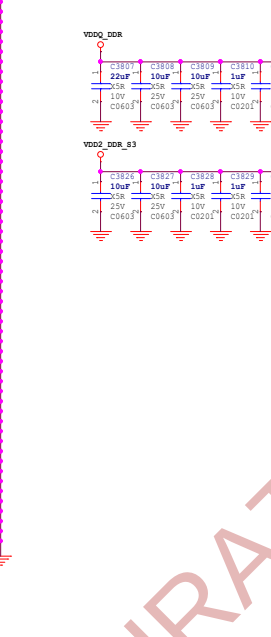
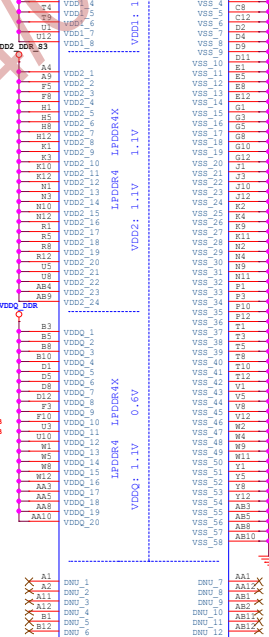
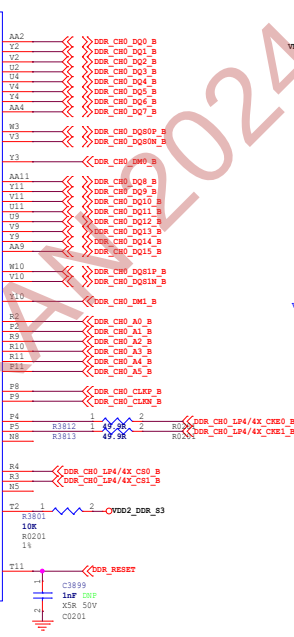
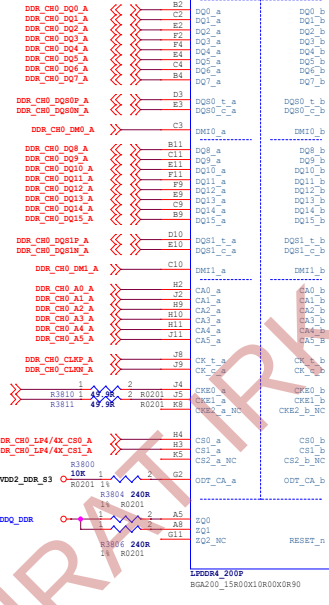
RTC_INT_L
I2C6_SCL_M3
I2C6_SDA_M3



Address:Read A3H,Write A2H

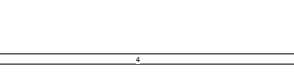
LPDDR4/4X

CH A CH B



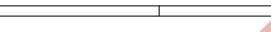
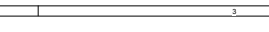
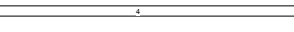
LPDDR4/4X

CH A CH B



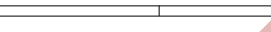
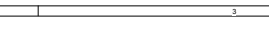
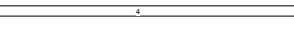
LPDDR4/4X

CH A CH B



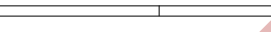
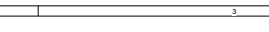
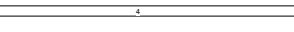
LPDDR4/4X

CH A CH B



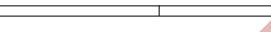
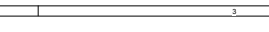
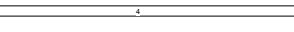
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CH A CH B



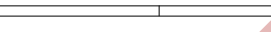
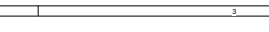
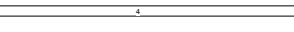
LPDDR4/4X

CH A CH B



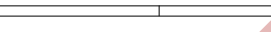
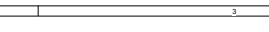
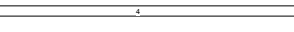
LPDDR4/4X

CH A CH B



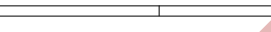
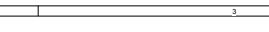
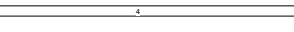
LPDDR4/4X

CH A CH B



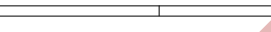
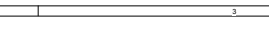
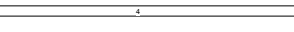
LPDDR4/4X

CH A CH B



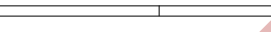
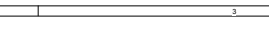
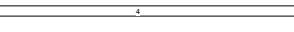
LPDDR4/4X

CH A CH B



LPDDR4/4X

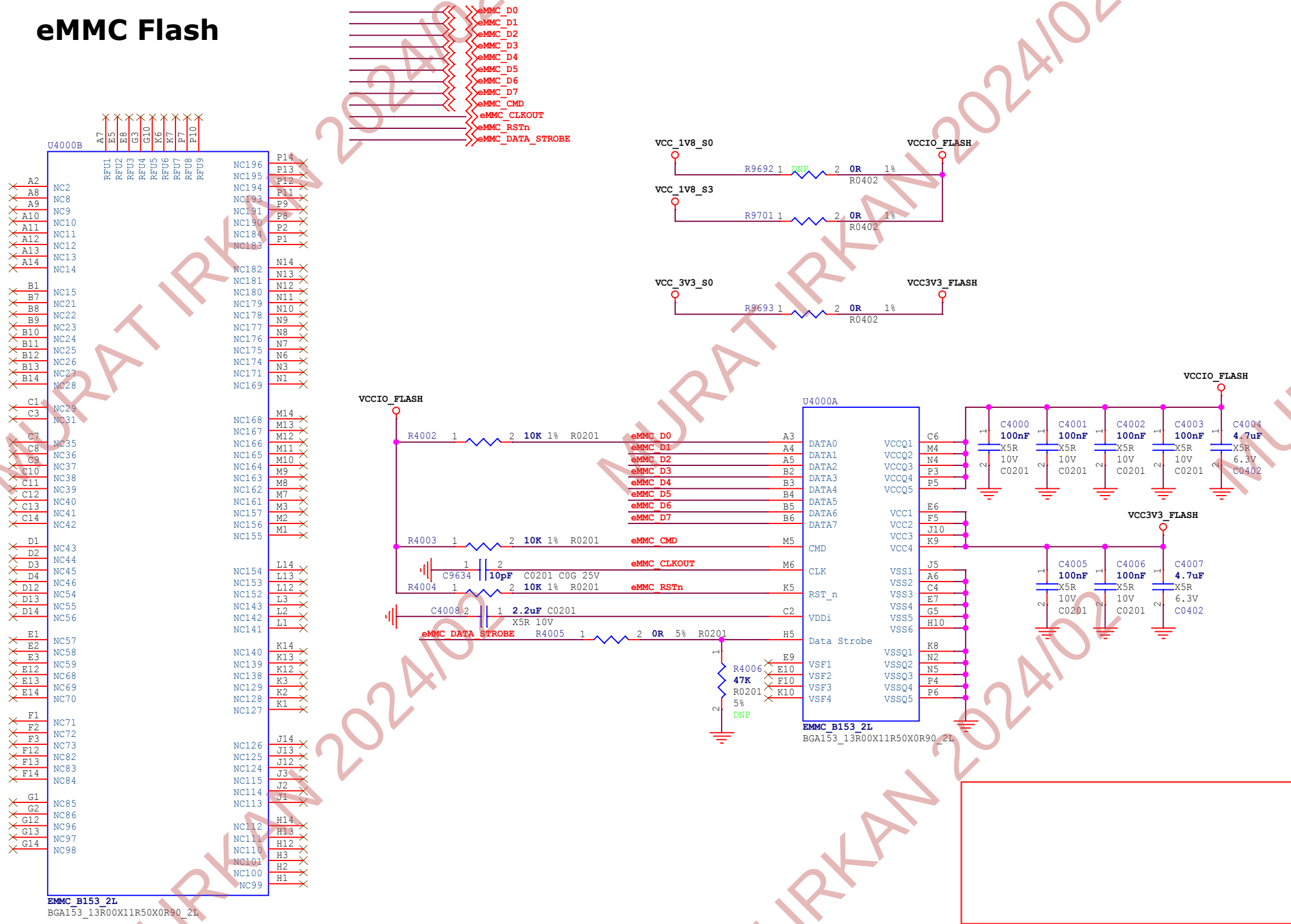
CH A CH B



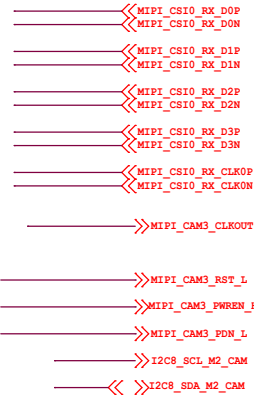
LPDDR4/4X

CH A CH B

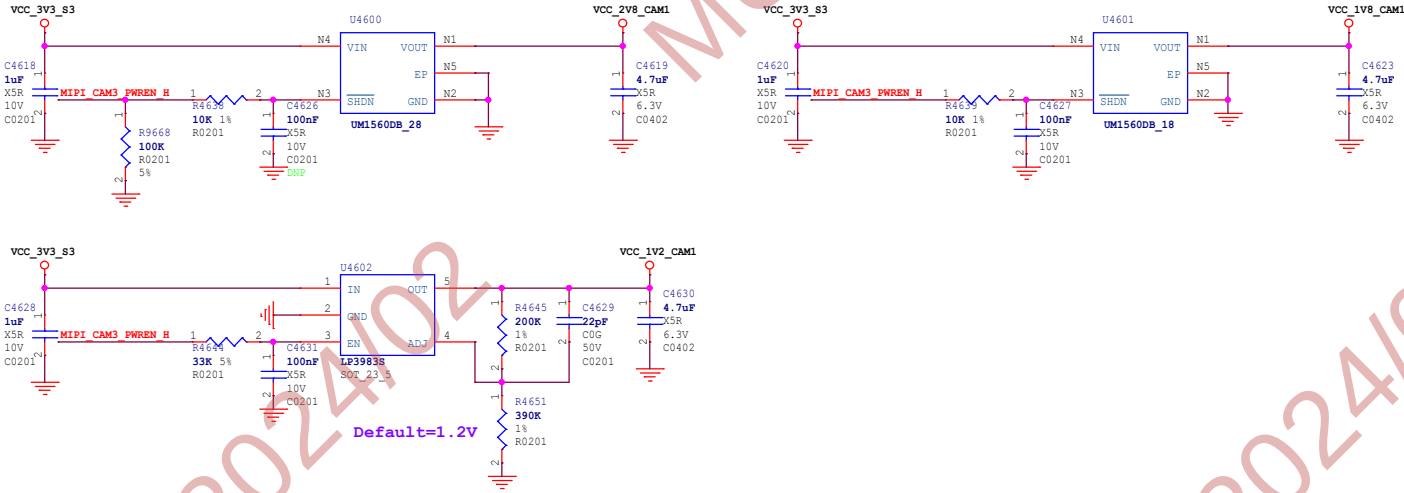
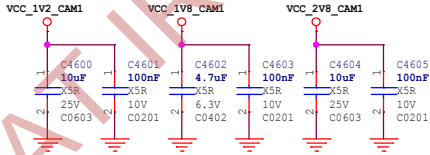
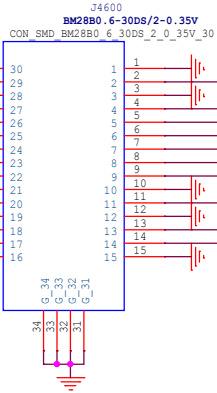
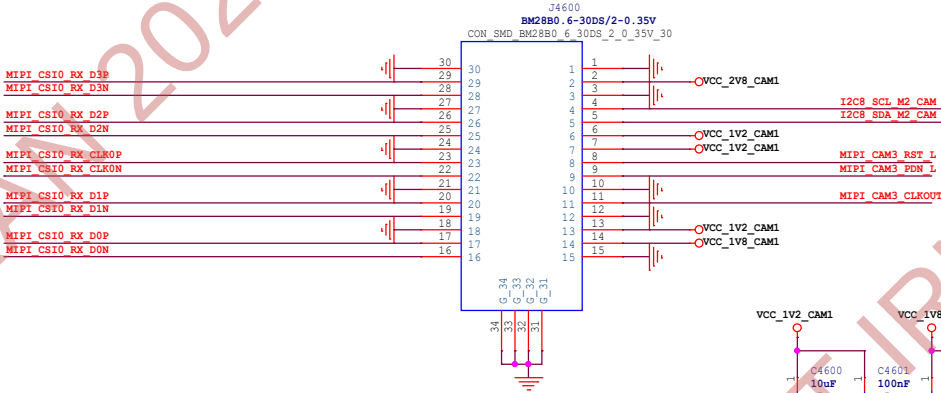
eMMC Flash



MIPI-CSIO_RX



前摄: OV16A10-GA5A-Z



VI-Camera DPHY_RX

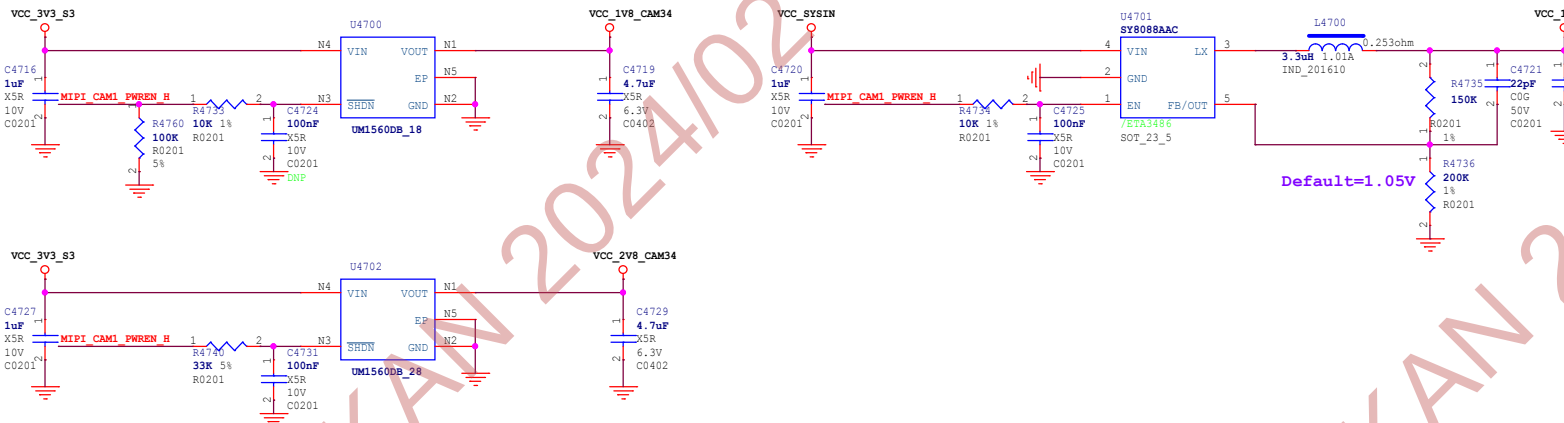
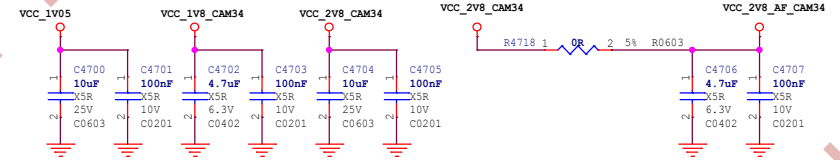
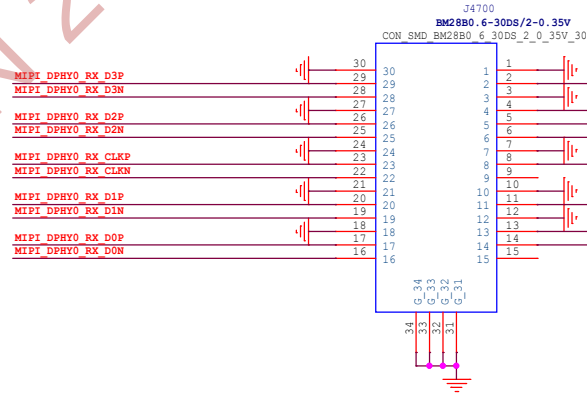
后摄: S5K3L6XX03-FGX9

MIIPI_DPHY0_RX_CLKP
MIIPI_DPHY0_RX_CLKN
MIIPI_DPHY0_RX_D0P
MIIPI_DPHY0_RX_D0N
MIIPI_DPHY0_RX_D1P
MIIPI_DPHY0_RX_D1N
MIIPI_DPHY0_RX_D2P
MIIPI_DPHY0_RX_D2N
MIIPI_DPHY0_RX_D3P
MIIPI_DPHY0_RX_D3N

MIIPI_CAM1_CLKROUT

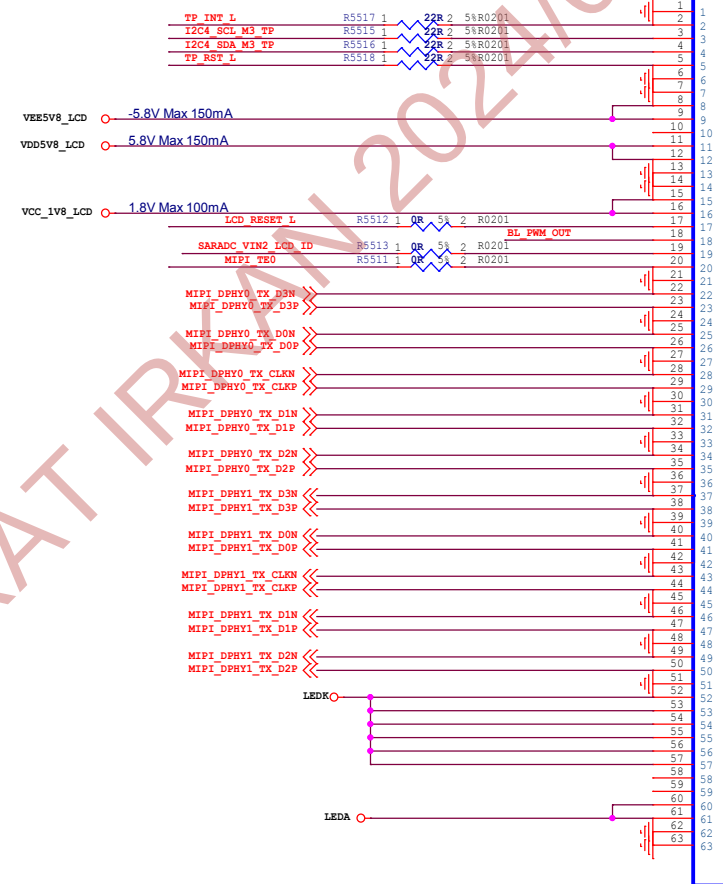
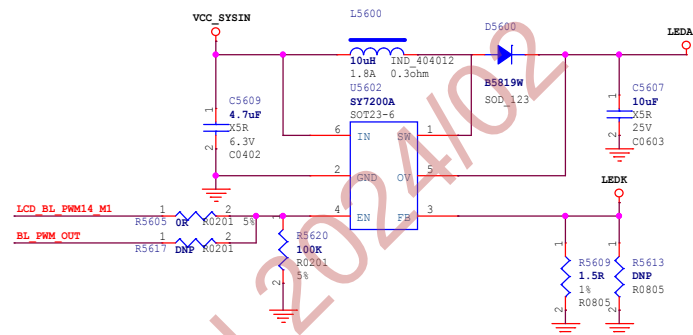
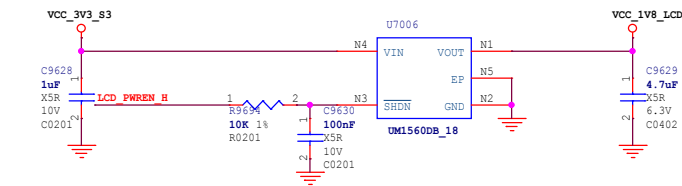
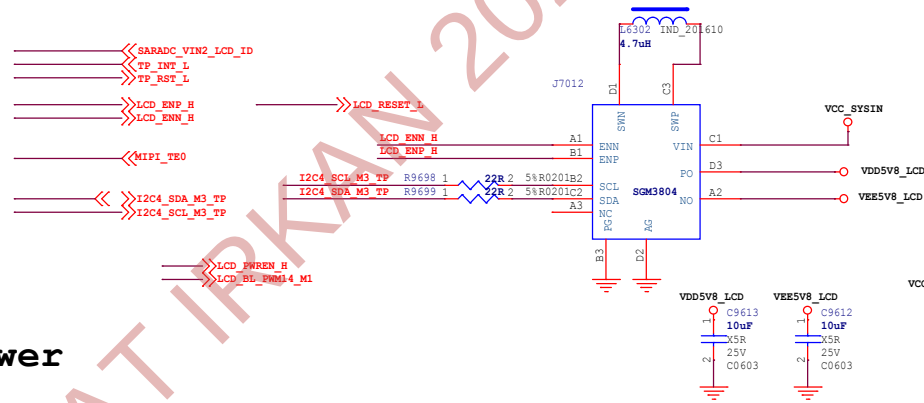
MIIPI_CAM1_PWREN_H
MIIPI_CAM1_RST_L

I2C5_SCL_M3_CAM
I2C5_SDA_M3_CAM



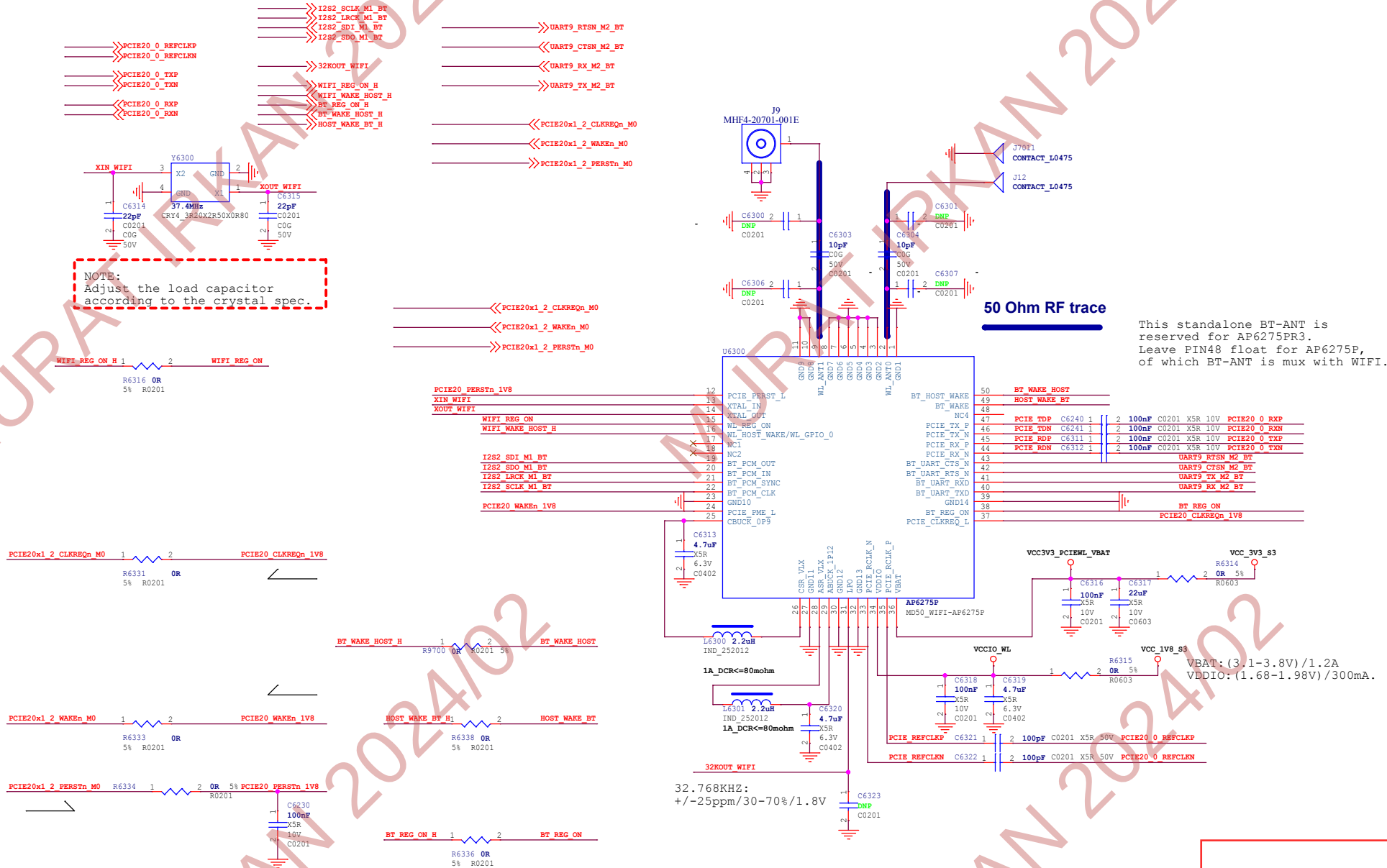
MIPI DPHY TX

Power

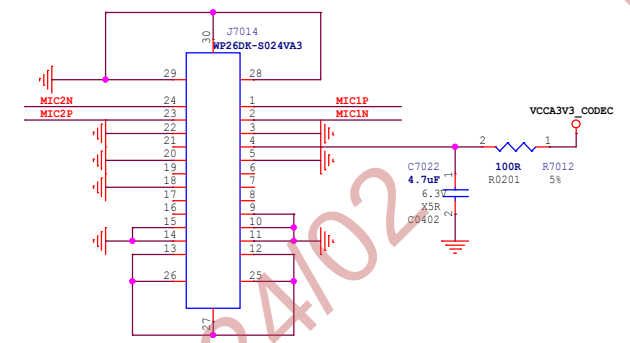
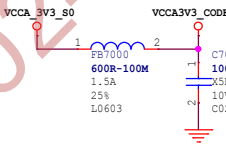


J5500
FH35C-61S-0.3SHW
CON SMD FH35C-61S-0.3SHW

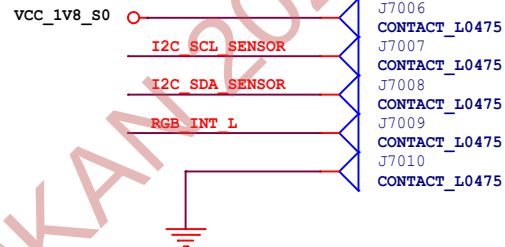
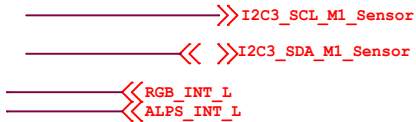
PCIE WIFI6/BT Module-2T2R



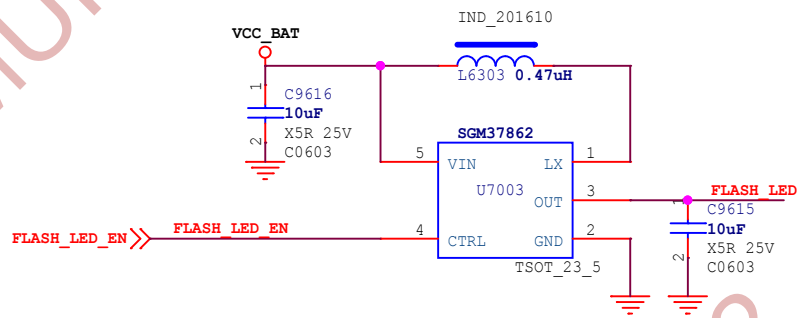
VCCA3V3_CODEC



Sensor



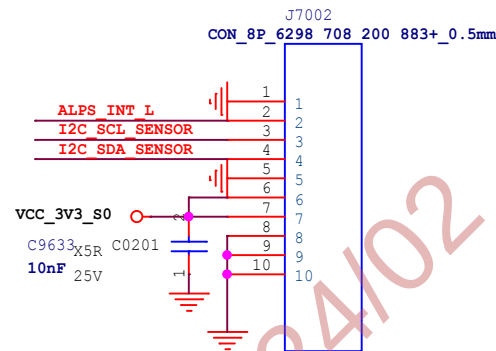
PLS+ALS



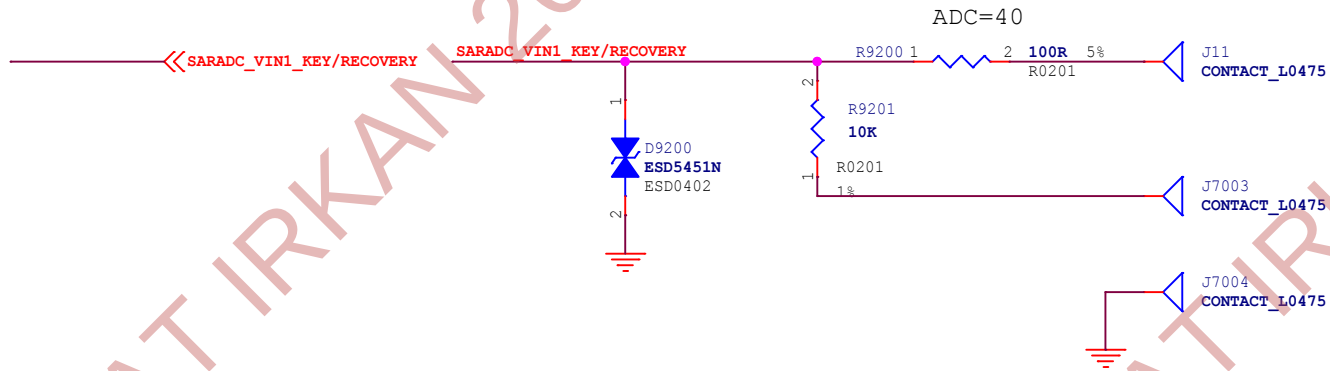
MAX: 1.5A



Flashlight



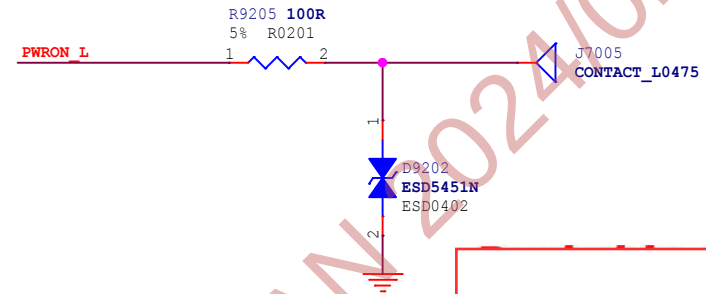
KEY Array



Reset_Key

RESET_L
PWRON_L

PWR_Key



UART Debug

JTAG Debug

UART2_TX_M0/JTAG_TCK_M2
UART2_RX_M0/JTAG_TMS_M2

UART2_TX_M0/JTAG_TCK_M2 R9300 1 2 22R R0201 5% UART_TX
UART2_RX_M0/JTAG_TMS_M2 R9301 1 2 22R R0201 5% UART_RX

